Power System Case Studies with PyPSA

Load flow, operating limits and renewable integration in a Sahel context

Nine case studies, one problem type each: solving the load flow, power balance, undervoltage, congestion and N-1 security, demand growth and voltage stability, reverse power flow and hosting capacity, and two real-data planning cases (Niamey grid with solar and storage, Agadez off-grid mini-grid).

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Code, datasets and executed notebooks: folder `pypsa-case-studies/`.
All numbers in this report are produced by that code (PyPSA 1.4, HiGHS, pvlib).
Grid parameters of the Niamey and Agadez cases are fictitious and educational;
weather data are real (NASA POWER, 2023).

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Executive summary

This report solves and explains nine power-system case studies with PyPSA. Each one isolates *one type of problem* that an engineer meets when studying a grid, states the exercise, builds a PyPSA dataset (buses, carriers, generators, loads, lines and their time series `loads_t`, `generators_t`), solves it, and interprets the result as an expert would present it to a jury or to decision-makers.

Case	Problem type	Main result
CS00	Non-linear equations	Newton converges quadratically to a simple root (8 digits in 5 steps) but only linearly to a double root (ratio 0.5); it fails where $f' = 0$.
CS01	Solving the AC load flow	3-bus Newton-Raphson from scratch: $\|U_3\| = 0.951$ p.u., 4 iterations; identical to PyPSA (10^{-15}). Literal exercise data are infeasible.
CS02	Power balance	30 MW of PV cut the thermal output from 50.3 to 20.2 MW and the losses by 31 %, but the evening peak stays on the thermal plant.
CS03	Undervoltage	Remote bus at 0.919 p.u., 24 h a day; a 15 MVAr capacitor bank restores 0.959 p.u. and cuts the daily losses by 16 %.
CS04	Congestion, N-1	Hot-season N-1 overload of 113 %; security-constrained OPF fixes it for 54 \$/h; the AC re-check reveals a voltage collapse (0.75 p.u.) needing 62 MVAr.
CS05	Demand growth, stability	Voltages look fine up to +30 % but all reactive reserve is gone; maximum loadability +77 % (vs a false +145 % if limits are ignored).
CS06	Reverse flow, hosting	Hosting capacity at the weak bus: 3 MW → 66 MW (lower set points) → 129 MW (PF 0.95); battery + reactive control remove all overvoltage.
CS07	Real case A, planning	Niamey, hourly 2023: without imports, 333 MW PV + 196 MW/783 MWh battery cut cost by 22 % and fuel by 68 %.
CS08	Real case B, energy access	Agadez mini-grid: PV + battery supply 94 % of demand; cost 0.30 → 0.16 \$/kWh, robust to fuel prices.

Five messages for decision-makers

1. Having enough megawatts does not mean the power reaches the customers: voltage and reactive power usually bind before line capacity (CS03, CS05, CS06).
2. The first warning of a stressed grid is the loss of reactive reserve, not a low voltage (CS05).
3. Cheap fixes come first: capacitor banks, voltage set points, inverter reactive control and preventive redispatch solve most problems at a fraction of the cost of new lines (CS03, CS04, CS06).
4. Solar is the cheapest energy in every case studied; storage is what lets the system absorb it (CS07, CS08).
5. Linear (DC) optimisation must always be checked with the full AC load flow (CS04, CS07).

1 Introduction

1.1 Purpose and approach

The load flow is the basic tool of power-system analysis: for a given generation, demand and network, it gives the steady-state voltages, flows and losses. The goal of this work is not only to obtain numbers with PyPSA, but to understand each electrical problem, to explain the results and to recommend actions, as an energy-systems engineer would. The case studies are organised by *problem type* rather than by network size (Table 1).

Table 1: Problem types, networks and methods.

Case	Problem type	Network	Method
CS00	Solving $f(x) = 0$	scalar	Newton's method
CS01	Solving the AC load flow	3 buses (Exercise 5)	Newton-Raphson from scratch, PyPSA `pf`
CS02	Power balance, slack, reactive power	3 buses, Niamey context	AC load flow, 24 h real profile
CS03	Undervoltage	5 buses, Niamey context	AC load flow, remedy comparison
CS04	Congestion, N-1 security, prices	IEEE 9-bus (MATPOWER)	DC-OPF, SC-OPF, AC N-1
CS05	Demand growth, voltage stability	IEEE 14-bus (MATPOWER, PGLib)	AC load flow, Q limits, PV curve
CS06	Reverse flow, hosting capacity, storage	IEEE 14-bus + PV + battery	AC load flow, 24 h real profile
CS07	Planning with variable renewables	5 buses, 8760 h, real weather	Capacity-expansion LOPF, AC checks
CS08	Off-grid sizing (energy access)	single bus, 8760 h, real weather	Capacity-expansion LOPF, sensitivity

1.2 Environment and repository

The work is reproducible with Python 3.11+ and the packages of `environment.yml`: PyPSA 1.4, the HiGHS solver, pvlib, pandas, matplotlib, jupytext and nbconvert.

```
conda env create -f environment.yml && conda activate pypsa-cases
python scripts/build_datasets.py      # rebuild the PyPSA CSV datasets
python scripts/run_all.py             # execute every case study -> notebooks/, results/
```

Folder	Content
`src/pscs/`	shared library: MATPOWER reader, Newton-Raphson, indicators, solar and demand models, chart style
`data/raw/`	official MATPOWER files, PGLib-OPF ratings, NASA POWER hourly weather, reference results
`data/networks/`	eight PyPSA datasets (CSV folders), each with a README
`scripts/`	one script per case study (source of the notebooks), dataset builder, runner
`notebooks/`	executed Jupyter notebooks with all outputs
`results/`	figures (PNG) and tables (CSV, JSON summaries)

1.3 Data sources

- **IEEE 9-bus and 14-bus systems**: official MATPOWER files `case9.m` and `case14.m`; thermal ratings of the 14-bus system from PGLib-OPF v23.07 (IEEE PES).
- **Weather**: NASA POWER hourly data for 2023 (CERES SYN1deg irradiance, MERRA-2 temperature and wind) for Niamey (13.51 N, 2.11 E) and Agadez (16.97 N, 7.99 E).

- **Renewables.ninja**: the code reads Renewables.ninja PV files directly and uses them automatically when present (`data/raw/renewables_ninja/`). The API could not be reached from the computing environment of this study, so the PV series were computed from NASA POWER with pvlib, using the same model chain as Renewables.ninja (section 2.5). A script, `scripts/fetch_renewables_ninja.py`, downloads the ninja files with a free API token; re-running the notebooks then uses them.
- **Teaching networks (CS02, CS03, CS07)**: fictitious parameters inspired by the Niamey context; they do not represent NIGELEC data.

2 Methods

2.1 AC load flow and Newton-Raphson

For every bus i, with $Y_{ik} = G_{ik} + jB_{ik}$ the admittance matrix and $\theta_{ik} = \theta_i - \theta_k$:

$$P_i = |V_i| \sum_k |V_k| \left(G_{ik} \cos\theta_{ik} + B_{ik} \sin\theta_{ik}\right), \tag{1}$$

$$Q_i = |V_i| \sum_k |V_k| \left(G_{ik} \sin\theta_{ik} - B_{ik} \cos\theta_{ik}\right). \tag{2}$$

Bus types fix two of the four quantities: slack ($|V|$, θ), PV (P, $|V|$), PQ (P, Q). The unknowns are θ at PV and PQ buses and $|V|$ at PQ buses. Newton-Raphson solves

$$\begin{bmatrix} \partial P/\partial\theta & \partial P/\partial|V| \\ \partial Q/\partial\theta & \partial Q/\partial|V| \end{bmatrix} \begin{bmatrix} \Delta\theta \\ \Delta|V| \end{bmatrix} = \begin{bmatrix} P^{spec} - P(\theta,|V|) \\ Q^{spec} - Q(\theta,|V|) \end{bmatrix} \tag{3}$$

from a flat start. Our own implementation (`src/pscs/newton.py`) is used to check PyPSA. On the official MATPOWER cases the two agree to machine precision and reproduce the published solutions (Table 2).

Table 2: Validation of the PyPSA models.

Case	Quantity	Reference	Deviation of PyPSA
Exercise 5 (CS01)	all voltages, flows	own Newton-Raphson	$< 4 \times 10^{-15}$
IEEE 9-bus	P_{G1} = 71.64 MW, losses 4.641 MW	own NR, MATPOWER solution	$< 10^{-13}$
IEEE 14-bus	slack 232.39 MW, losses 13.393 MW	own NR, pandapower	$< 10^{-13}$
IEEE 14-bus	bus voltages	IEEE CDF stored solution	0.0013 p.u. (rounding)

2.2 The PyPSA data model and datasets

A PyPSA network is a set of component tables. Static data sit in `n.buses`, `n.generators`, `n.loads`, `n.lines`, `n.transformers`, `n.storage_units`, `n.carriers`; time series in `n.loads_t.p_set`, `n.generators_t.p_max_pu`, `n.generators_t.p_set`, etc. Each dataset of this study is a standard CSV folder that any PyPSA user can open:

```
import pypsa
n = pypsa.Network("data/networks/niamey_3bus")   # buses.csv, lines.csv, loads-p_set.csv ...
n.pf()                                           # one AC load flow per snapshot (24 h)
n.buses_t.v_mag_pu                               # voltages, p.u.
n.lines_t.p0                                     # active flow at the sending end, MW
```

Unit conversion. PyPSA stores line impedances in ohm and admittances in siemens. With $Z_b = V_{nom}^2/S_b$: $r_\Omega = r_{pu} Z_b$, $x_\Omega = x_{pu} Z_b$, $b_S = b_{pu}/Z_b$. Transformers keep per-unit values on their own rating s_{nom}. Unit errors are the most frequent mistake in load-flow studies; the validation of Table 2 protects against them.

2.3 Reactive limits: PV to PQ switching

PyPSA's load flow holds the voltage of PV buses whatever reactive power it takes. Real generators cannot. We apply the standard iteration (`pscs.kpi.enforce_q_limits`): after each load flow, a generator above Q_{max} (below Q_{min}) becomes a PQ injection fixed at its limit, and the load flow is repeated until no limit is violated.

```
for g in pv_generators:
    if q[g] > q_max[g]:
        n.generators.loc[g, "control"] = "PQ"     # loses voltage control
        n.generators.loc[g, "q_set"] = q_max[g]   # stuck at its limit
```

2.4 Optimisation: DC-OPF, security-constrained OPF, capacity expansion

`n.optimize()` solves a linear (or quadratic) optimal power flow over all snapshots:

$$\min \sum_a c_a^{cap} \bar{P}_a + \sum_t \sum_g \left(c_{1,g}\, p_{g,t} + c_{2,g}\, p_{g,t}^2 \right) \tag{4}$$

subject to nodal balance, linearised flows $|P_{\ell,t}| \le s_{max,pu}\, s_{nom,\ell}$, availability $p_{g,t} \le \bar{P}_g\, p_{max,pu,g,t}$ and storage dynamics. For capacity expansion, c^{cap} is the annuity of the investment plus fixed O&M: $c^{cap} = \text{CAPEX} \cdot \left(\frac{r}{1-(1+r)^{-L}} + \text{FOM} \right)$ with $r = 8\,\%$. `n.optimize.optimize_security_constrained()` adds, for each outage c, the post-contingency flows $P_\ell + \text{LODF}_{\ell c} P_c$. The dual of the nodal balance is the locational marginal price.

2.5 Solar model chain

PV output per kWp is computed with pvlib from NASA POWER hourly data: solar position at mid-hour, Hay-Davies transposition to a south-facing plane tilted at the latitude, Faiman cell temperature, PVWatts DC model ($-0.4\,\%/°\text{C}$) and 10 % system losses (the Renewables.ninja default). Results for 2023: Niamey 19.5 % capacity factor (1 710 kWh/kWp, lowest in August during the monsoon), Agadez 21.2 % (1 857 kWh/kWp).

2.6 Demand model

No public hourly load series of the Niamey grid exists, so demand is *modelled* with the real temperature: $D(t) = k\, s(h)\, [1 + 0.025 \max(T_{24}(t) - 25°\text{C}, 0)]$, with a daily shape $s(h)$ peaking at 19:00-20:00 (lighting, households, cooling) and T_{24} the 24-hour mean temperature. The annual peak falls on 4 May 2023 at 19:00, in the hot season. This is an explicit assumption, stated wherever demand matters.

3 CS00 - Newton's method

Problem. Solve $f(x) = 10x^2 - 2x^3 = 0$ by Newton's method (Exercise 5, Assignment 1). $f'(x) = 20x - 6x^2$; roots $x = 0$ (double) and $x = 5$ (simple); f' vanishes at 0 and $10/3$.

Interpretation. From $x_0 = 1$ the method converges to the double root, but only linearly: $x_{k+1} = x_k(10 - 4x_k)/(20 - 6x_k) \to x_k/2$, so the error halves at each step. From $x_0 = 4$ it first overshoots to 6, then converges quadratically to 5 (8 exact digits after 5 iterations). At $x_0 = 10/3$ the tangent is horizontal and the method fails; about half of the starting points in $[-2, 8]$ end at each root. *Link with load flows:* the Jacobian replaces f'; a flat start is a good guess; several solutions may exist; a singular Jacobian marks the voltage-collapse point (CS05).

Table 3: Five Newton iterations from $x_0 = 1$ and $x_0 = 4$.

	(a) $x_0 = 1$				(b) $x_0 = 4$			
k	x_k	$f(x_k)$	$f'(x_k)$	Δx	x_k	$f(x_k)$	$f'(x_k)$	Δx
0	1.00000	8.00000	14.0000	−0.57143	4.00000	32.0000	−16.0000	2.00000
1	0.42857	1.67930	7.46939	−0.22482	6.00000	−72.0000	−96.0000	−0.75000
2	0.20375	0.39821	3.82586	−0.10408	5.25000	−13.7813	−60.3750	−0.22826
3	0.09966	0.09735	1.93366	−0.05034	5.02174	−1.09643	−50.8724	−0.02155
4	0.04932	0.02408	0.97179	−0.02478	5.00019	−0.00933	−50.0075	−0.00019
5	0.02454				5.00000			

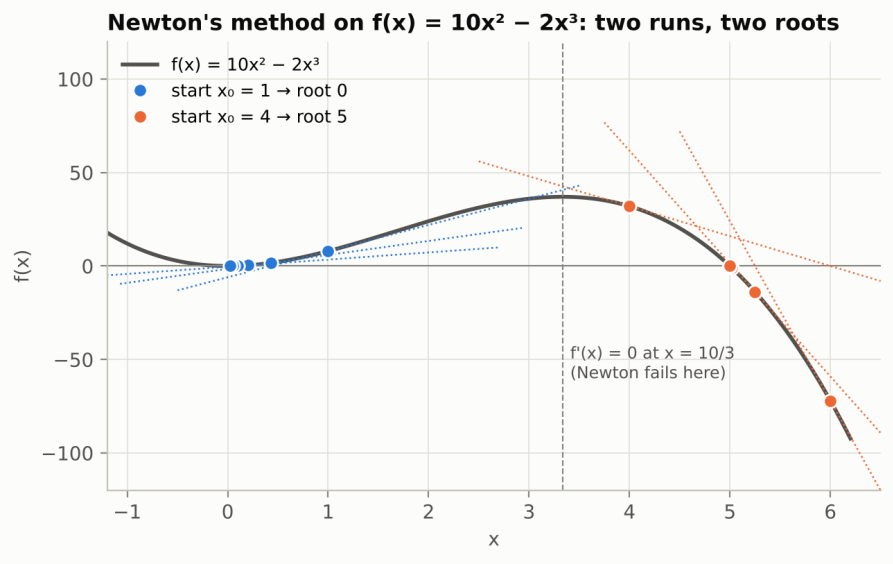


Figure 1: $f(x)$, the iterates and the first tangents of each run.

4 CS01 - Solving the AC load flow (Exercise 5)

Problem. Three buses and three identical pi lines; bus 1 slack ($|U_1| = 1.00$), bus 2 PV ($P_{G2} = 0.50$, $|U_2| = 1.05$), bus 3 PQ ($P_{D3} = 2.50$, $Q_{D3} = 1.00$ p.u.).

Data check. The exercise gives $\underline{Y}_{ik} = j0.1$ and $\underline{Y}^0_{ik} = -j8$. Read literally, the series element would be a capacitor and the shunt an 8 p.u. reactor; the Newton iteration diverges and, physically, a series admittance of 0.1 p.u. cannot carry 2.5 p.u. We use the swapped values (series $-j8$, total shunt $j0.1$, i.e. $j0.05$ per end); reading both values as impedances gives the same qualitative answer ($|U_3| = 0.971$).

Admittance matrix. $Y_{ii} = 2(-j8) + 2(j0.05) = -j15.9$ and $Y_{ik} = +j8$.

Load-flow problem. Unknowns $\theta_2, \theta_3, |U_3|$; equations $P_2 = 0.5$, $P_3 = -2.5$, $Q_3 = -1.0$ (no Q_2 equation: PV bus). Jacobian at the flat start:

$$J_0 = \begin{bmatrix} 16.8 & -8.4 & 0 \\ -8.4 & 16.4 & 0 \\ 0 & 0 & 15.4 \end{bmatrix}.$$

Table 4: Newton-Raphson iterations (Exercise 5).

Iteration	max |mismatch| (p.u.)	θ_2 (°)	θ_3 (°)	$\|U_3\|$ (p.u.)
0	2.5	0.0000	0.0000	1.00000
1	2.08×10^{-1}	−3.5782	−10.5668	0.96753
2	4.13×10^{-3}	−3.6522	−11.0830	0.95182
3	1.85×10^{-6}	−3.6530	−11.0949	0.95149
4	3.9×10^{-13}	−3.6530	−11.0949	0.95149

Results. $|U_3| = 0.951$ p.u., $\theta_2 = -3.65°$, $\theta_3 = -11.09°$; $P_{G1} = 2.000$ p.u. (lossless lines: $P_{G1} = P_{D3} - P_{G2}$), $Q_{G1} = 0.047$ p.u., $Q_{G2} = 1.222$ p.u. Total losses $S_{v,ges} = 0 + j0.269$ p.u.: the series reactances consume more reactive power than the line capacitances produce. PyPSA gives the same values to 4×10^{-15}.

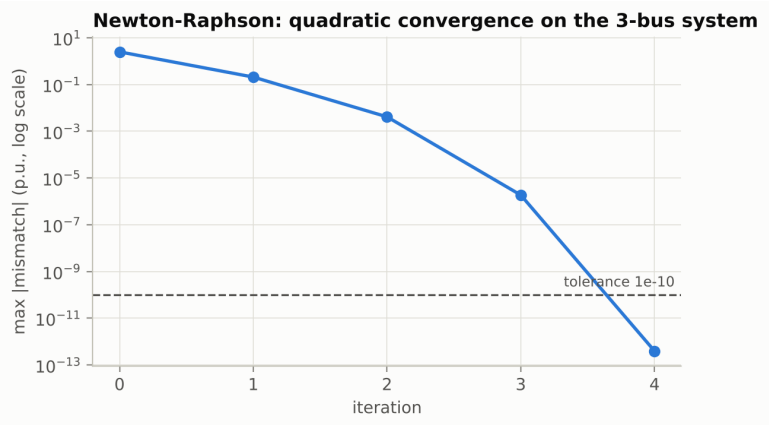


Figure 2: Quadratic convergence: the number of correct digits doubles at each iteration.

5 CS02 - Power sharing (3-bus, Niamey context)

Problem. A thermal plant (slack, 1.02 p.u.), a city (50 MW + 20 MVAr at the peak) and a 30 MW PV plant, linked by 15-25 km 132 kV lines (100 MVA hot-season rating).

Answers. (1) Voltages are close to nominal and within 0.95-1.05 p.u. (2) PV cuts the thermal output from 50.3 to 20.2 MW: the estimate $50 - 30 = 20$ MW is right to within the losses. (3) At unity power factor the thermal plant supplies the reactive power; in voltage-control mode the inverters give 11.7 MVAr, within the

Table 5: Design point (city at its peak, PV at 30 MW).

Variant	$\lvert V_2 \rvert$	$\lvert V_3 \rvert$	thermal P (MW)	thermal Q (MVAr)	PV Q (MVAr)	losses (MW)
No PV	1.009	1.014	50.28	17.8	0.0	0.280
PV, unity power factor	1.011	1.017	20.19	17.5	0.0	0.192
PV, voltage control (1.02)	1.013	1.020	20.18	5.7	11.7	0.180

13.7 MVAr capability of 33 MVA inverters at full output. (4) Maximum loading 33 %. (5) Losses fall from 0.28 to 0.19 MW.

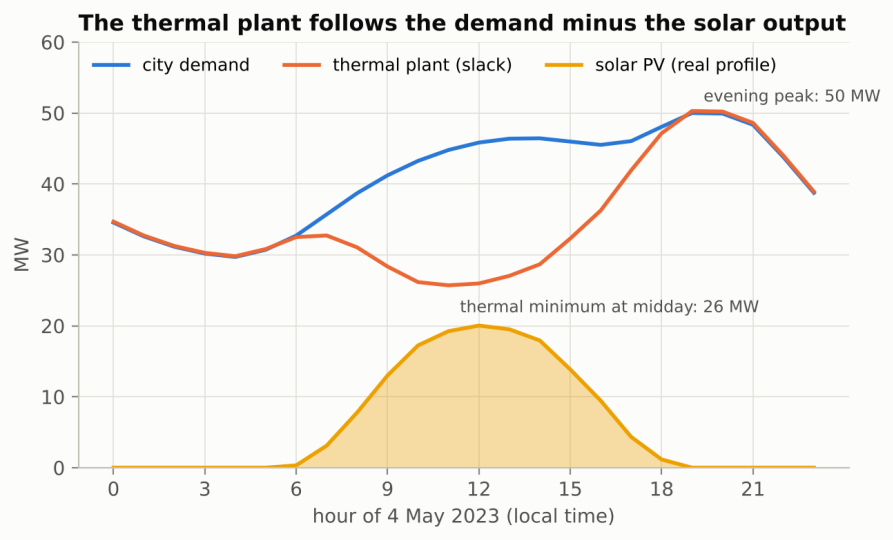


Figure 3: Real day (4 May 2023, annual peak): the thermal plant follows demand minus solar.

Real day. On the peak day the PV plant produces at most 20 MW (hot, hazy sky) and 15 % of the daily energy. The peak occurs after sunset, so the thermal plant must still be sized for 50 MW and ramp steeply between 16:00 and 19:00. Solar saves fuel; without storage it does not reduce the firm capacity needed.

6 CS03 - Undervoltage at a remote industrial zone

Problem. Five buses; the industrial zone (35 MW + 18 MVAr) is at the end of an 85 km line. The exercise *assumed* 0.92 p.u.; we compute it.

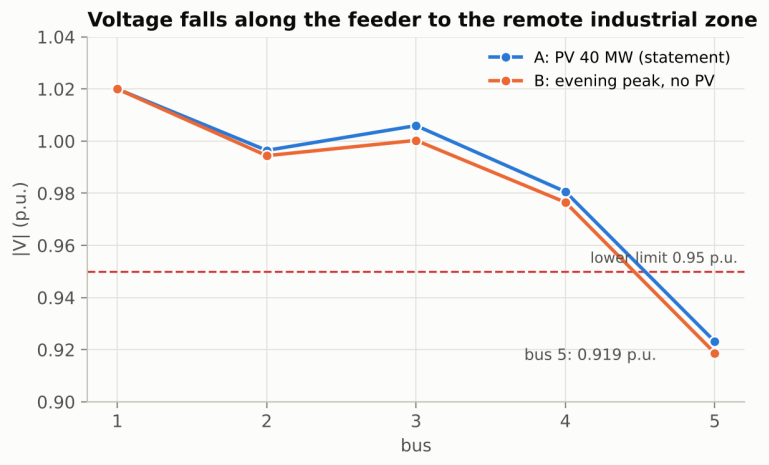


Figure 4: Voltage profile: only bus 5 violates the limit; the worst case is the evening peak.

$V_5 = 0.923$ p.u. with PV at 40 MW and 0.919 p.u. at the evening peak without sun: the PV plant, far from bus 5, hardly helps. On an 85 km line $X = 33\,\Omega$ and $R = 10\,\Omega$: reactive power costs three times more voltage than active power ($\Delta V \approx (RP + XQ)/V$).

Table 6: Remedies at the evening peak.

Remedy	$\lvert V_5 \rvert$ (p.u.)	losses (MW)	assessment
None	0.919	3.18	violates
Capacitor bank 15 MVAr at bus 5	0.959	2.72	cheapest, fixes the cause
Plant set point 1.05 p.u.	0.953	2.96	uses up the upper margin
Second circuit on line 4-5	0.956	2.48	best losses, most expensive
PV inverters Q-at-night (1.03 p.u.)	0.942	3.22	needs 68 MVAr $>$ 44 MVA: infeasible
SVC/STATCOM at bus 5 (0.97 p.u.)	0.970	2.66	17.9 MVAr, precise, costlier

The minimum bank is 11.8 MVAr (sensitivity 0.28 % of voltage per MVAr); a 15 MVAr bank in three 5 MVAr steps is recommended. Over the real day, bus 5 is below 0.95 p.u. 24 hours out of 24 without action and never with the bank, with no overvoltage at night; daily losses fall from 59.7 to 50.1 MWh.

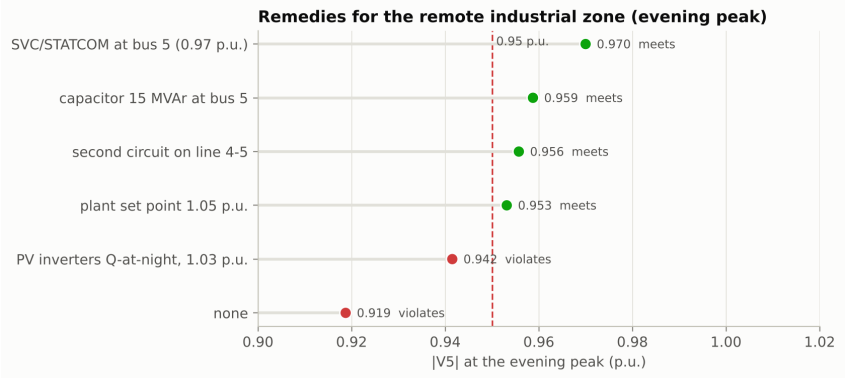


Figure 5: Comparison of the remedies at the evening peak.

7 CS04 - Congestion and N-1 security (IEEE 9-bus)

Problem. Hot-season ratings (15 % below MATPOWER values, an assumption) and an evening peak 15 % above the base case. Generator costs are the quadratic MATPOWER costs.

Economic dispatch. G1 101.4 MW, G2 153.5 MW, G3 107.3 MW, 6 429 $/h, uniform nodal price 27.30 $/MWh. The intact grid is fine (maximum loading 73 %).

N-1 screening (AC). The loss of line 8-9 overloads line 5-6 to 113 % of its rating: the dispatch is not N-1 secure. The loss of line 9-4 leaves bus 9 at 0.75 p.u.

Preventive redispatch. The security-constrained OPF moves 18 MW from G2 and G3 to G1: G1 119.8, G2 142.7, G3 99.8 MW, for 54 $/h (0.84 %). Nodal prices separate (25.5 to 31.3 $/MWh): the spread is the value of transport capacity. The AC re-check confirms 98.7 % on line 5-6, but flags line 6-7 at 106 % after the loss of line 9-4: this is the voltage collapse at bus 9, invisible to the DC model. 62 MVAr of reactive support switched in at bus 9 after that outage restores 0.95 p.u. and brings line 6-7 back to 98.5 %.

Redispatch or reinforce? Uprating line 5-6 to 200 MVA makes the economic dispatch N-1 secure for flows (worst 97 %) but does not cure bus 9. At 600 critical hours a year, redispatch costs about 32 k$/year: reinforcement is justified only if congestion becomes frequent; the price spread tells the planner when.

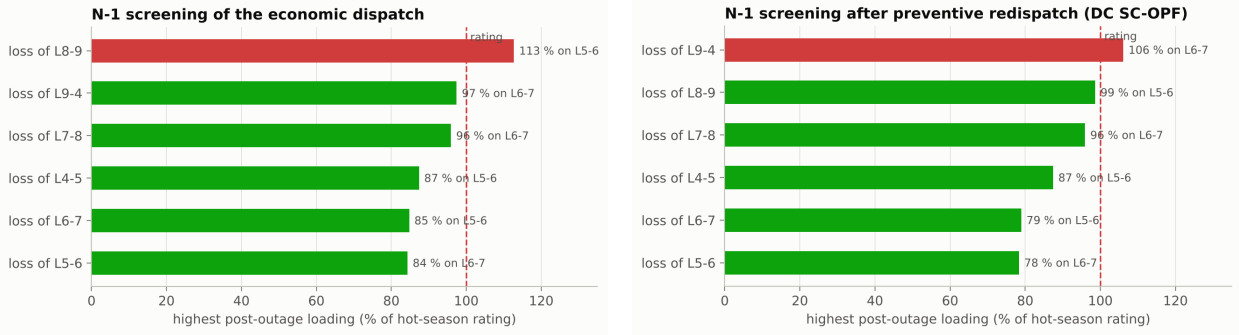


Figure 6: Worst post-outage loading for each line outage: economic dispatch (left) and after the security-constrained OPF (right).

8 CS05 - Demand growth and voltage stability (IEEE 14-bus)

Problem. All loads (and G2) grow by 10 to 50 %; generator reactive limits are enforced.

Table 7: Indicators per scenario (reactive limits enforced).

Scenario	min $\|V\|$ (p.u.)	buses < 0.95	max loading (%)	losses (MW)	gen Q (MVAr)	units at Q limit
S0 100 %	1.010	0	59.1	13.4	82	0 of 4
S1 110 %	1.010	0	65.3	16.4	102	1 of 4
S2 120 %	1.006	0	71.7	19.8	123	3 of 4
S3 130 %	0.983	0	78.7	23.9	149	4 of 4
S4 140 %	0.954	0	86.6	28.9	179	4 of 4
S5 150 %	0.909	9	95.5	35.2	215	4 of 4

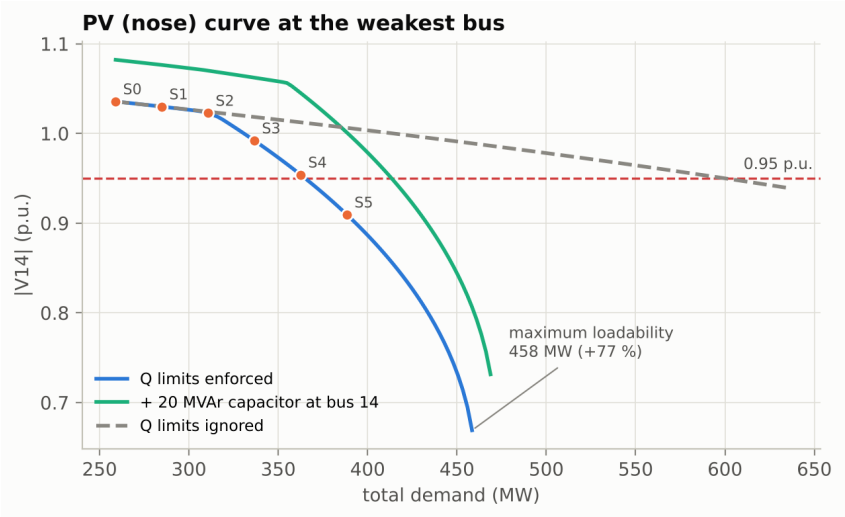


Figure 7: PV (nose) curve at bus 14: maximum loadability +77 % with reactive limits.

Interpretation. Up to +30 % the classic indicators look healthy, yet all four voltage-controlling units reach their reactive limit: the reserve is exhausted. Beyond, voltages fall fast (nine buses below 0.95 p.u. at +50 %) and losses grow with the square of the current (13.4 to 35.2 MW). The maximum loadability is +77 % (458 MW); a 20 MVAr capacitor at bus 14 raises it to +81 %. Ignoring the reactive limits (PyPSA default) would wrongly suggest more than +145 %. *Recommendations:* reactive compensation at weak buses, voltage-control capability on new plants, reinforcement of corridor 1-5, peak shaving.

9 CS06 - Reverse power flow, hosting capacity and storage

Problem. A PV plant at bus 14 of the IEEE 14-bus system (weakest node, local load about 14 MW at noon), real solar profile of 16 March 2023; scenarios S0 to S4 of the exercise.

Table 8: Hosting capacity of bus 14 at the hour of maximum sun.

Strategy	Hosting capacity	First limit
Unity PF, original IEEE voltage set points	3 MW	overvoltage
Unity PF, generator set points −0.03 p.u.	66 MW	overvoltage
PF 0.95 absorbing, set points −0.03 p.u.	129 MW	undervoltage

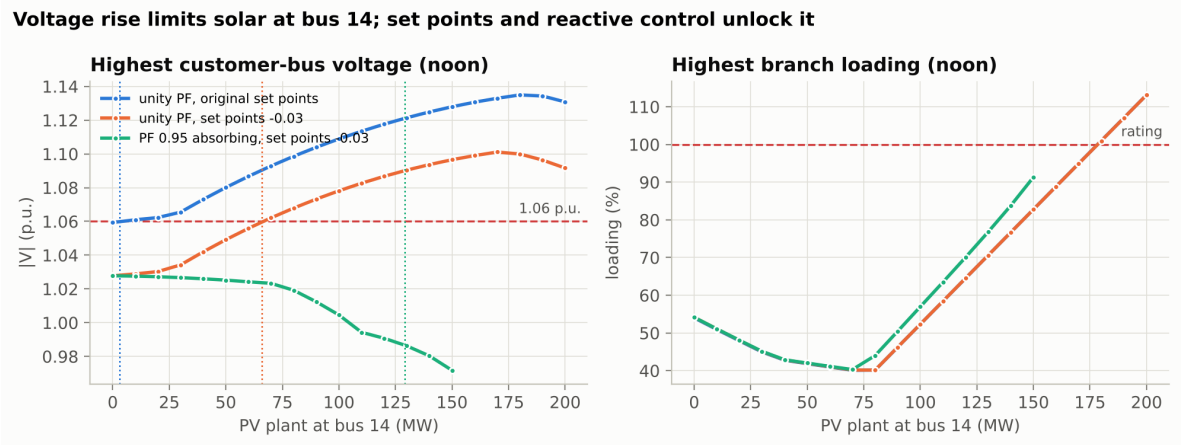


Figure 8: Voltage rise, not line capacity, limits solar at bus 14.

Table 9: Twenty-four hour scenarios.

Scenario	max $\lvert V_{14}\rvert$	overvoltage hours	reverse-flow hours	max export (MW)	losses (MWh)
S0 no PV	1.057	15	0	0.0	211.0
S1 PV 50 MW	1.073	20	9	25.2	186.2
S2 PV 100 MW	1.098	20	10	62.4	180.2
S3 S2 + battery 50 MW/200 MWh	1.080	24	14	37.5	166.6
S4 S3 + PF 0.95 + set points	1.046	0	14	37.4	183.8

Interpretation. The voltage plan comes first: with the original set points the customer buses already exceed 1.06 p.u. at low load. Lowering the set points (free) and absorbing reactive power multiply the hosting capacity by more than 40. The flow on lines 9-14 and 13-14 reverses as soon as PV exceeds the local load. The battery (peak shaving) lowers the peak voltage and the export, but its evening discharge at the weak bus raises the voltage there: storage needs a voltage-aware dispatch. The full package (S4) removes every overvoltage hour.

10 CS07 - Real case A: solar and storage for the Niamey area

Question. How much PV and storage minimises the yearly cost of supply, with and without imports, and what does it change for fuel, CO_2 and the grid? Hourly year 2023 (real weather, modelled demand: 726 GWh, peak 105 MW), five-bus network of CS03 with the 15 MVAr bank. Costs: PV 800 \$/kW (annualised 86.9 k\$/MW/yr), 4-hour battery 1 150 \$/kW (157.4 k\$/MW/yr), thermal fuel 180 \$/MWh, imports 70 \$/MWh. When imports are interrupted the grid is an island and the thermal plant keeps 20 MW online (must-run).

Interpretation. Solar costs about 51 \$/MWh when fully used, far below fuel. Without storage the optimum stops at 154 MW with 20 % curtailment: the midday demand minus the must-run cannot absorb more. *This,*

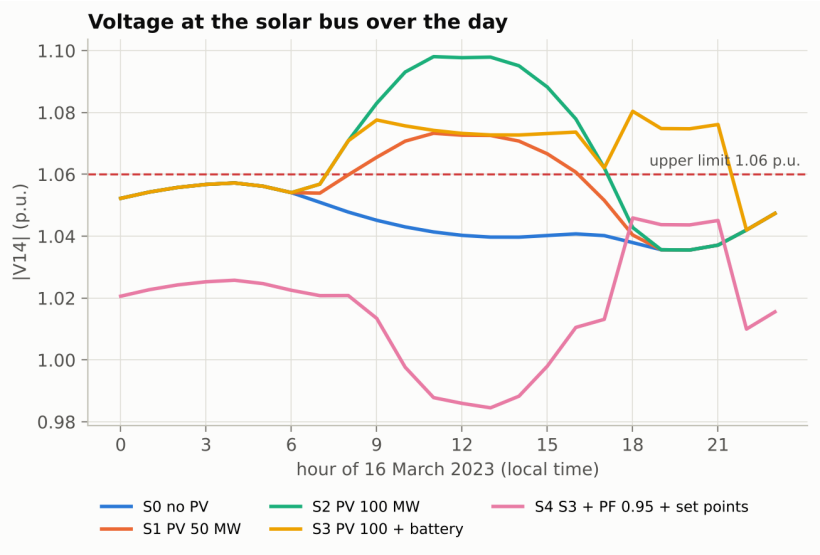


Figure 9: Voltage at bus 14 over the day for the five scenarios.

Table 10: Optimisation results (8760 hours of 2023).

Scenario	PV (MW)	battery (MW / MWh)	solar share	CO_2 (kt)	cost (M$/yr)	$/MWh
A imports, no PV	0	0 / 0	0 %	35	56.4	77.6
B no imports, no PV	0	0 / 0	0 %	508	130.7	180.0
C no imports, 100 MW PV	100	0 / 0	23 %	391	109.3	150.5
D0 no imports, optimal PV	154	0 / 0	29 %	361	106.2	146.2
D no imports, optimal PV + battery	333	196 / 783	72 %	163	101.7	140.1
E imports, optimal PV + battery	144	8 / 33	33 %	7	49.0	67.5

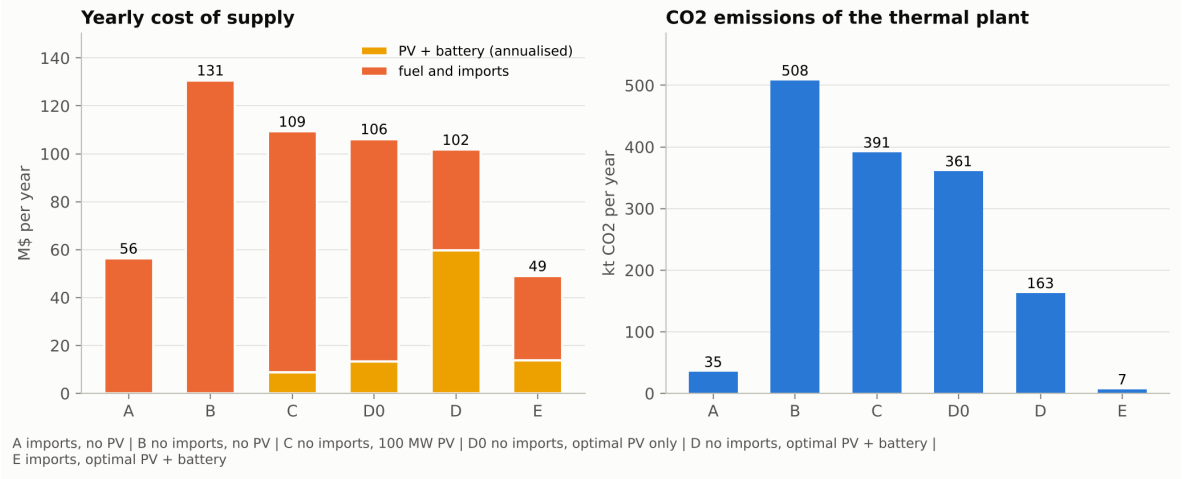


Figure 10: Yearly cost and CO_2 emissions of the six scenarios.

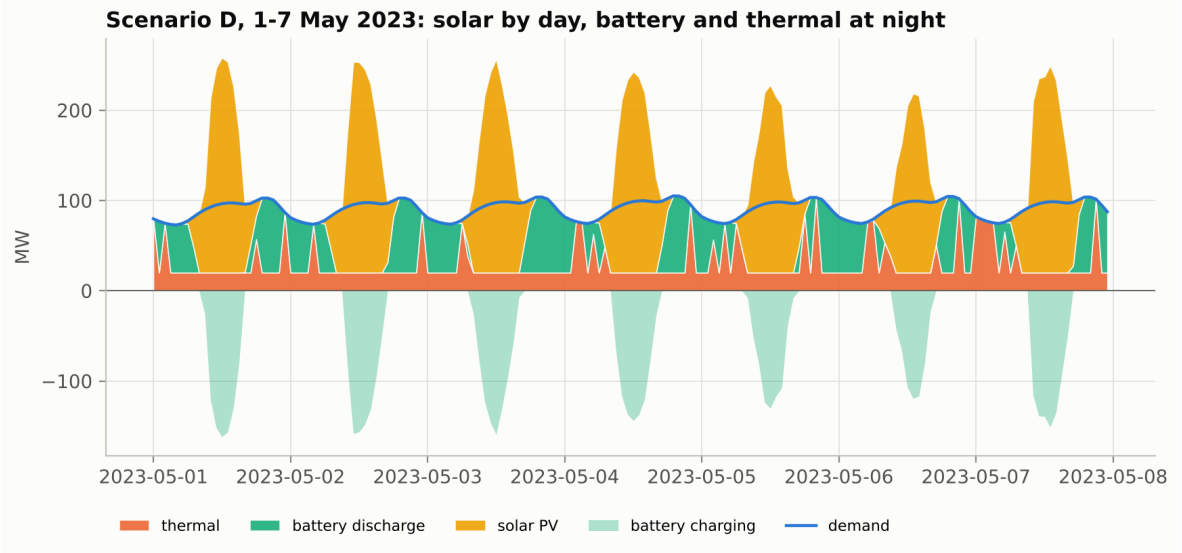


Figure 11: Scenario D in the hot season: solar by day, battery in the evening, thermal must-run at night.

not the grid, limits solar: lines never exceed about 70 % of their rating. Storage moves the limit: 333 MW of PV and a 196 MW / 783 MWh battery supply 72 % of the demand, cut fuel by 68 % and the yearly cost by 22 % compared with B. With imports, cheap night imports leave little room for storage (E). The AC load flow of scenario D at the evening peak and at the solar peak gives voltages of 0.968-1.02 p.u. and loadings below 70 %: the plan is technically feasible. Monsoon weeks show that the thermal plant remains necessary for security of supply.

11 CS08 - Real case B: Agadez off-grid mini-grid

Question. Least-cost mix of PV and batteries for a diesel-supplied town (12.4 GWh/yr, peak 2 MW), real Agadez weather 2023, remote-site costs (PV 1 000 $/kW, battery 350 $/kWh + 200 $/kW), diesel 1.00 $/litre at 0.28 l/kWh.

Table 11: Diesel-only versus optimal hybrid (diesel at 1.00 $/litre).

	Diesel only	Optimal hybrid
PV / battery	–	7.5 MW / 4.5 MW, 17.9 MWh
Renewable share	0 %	94 %
Diesel (litres/yr)	3.48 million	0.22 million
Genset running hours	8 760	773
CO_2 (t/yr)	9 209	570
Cost of electricity ($/kWh)	0.300	0.163

Interpretation. The hybrid halves the cost of electricity and cuts fuel by 94 %. The result is robust: at 0.8 $/litre the hybrid still costs 0.159 $/kWh against 0.244; at higher prices the optimiser simply builds more PV and storage. The genset remains as back-up for cloudy monsoon days. Priorities: concessional finance for the up-front cost, a cost-reflective tariff, local maintenance capacity.

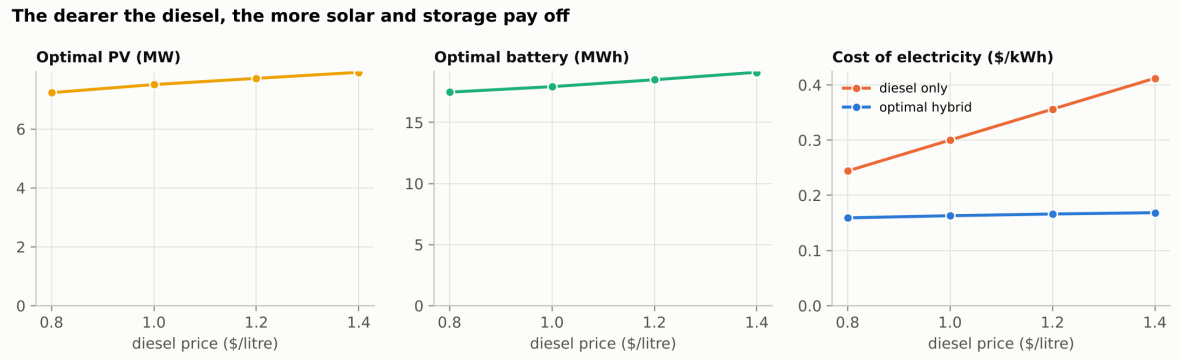


Figure 12: Optimal sizes and cost of electricity for diesel prices from 0.8 to 1.4 $/litre.

12 Cross-cutting lessons

Problem	First signs	Remedies, cheapest first
Undervoltage	low $\|V\|$ at remote PQ buses, high reactive flows	local capacitors, set points, SVC/STATCOM, new circuit
Congestion	loading > rating, N-1 violations, price spread	redispatch, SC-OPF, reconductoring, new line
Load growth	generators at Q limits, rising losses	reactive compensation, local generation, reinforcement, demand response
Solar integration	voltage rise, reverse flow, curtailment	voltage plan, inverter reactive control, storage, curtailment
Variability	midday surplus, evening peak	storage, flexible thermal, imports, demand shifting

PyPSA lessons from this work (version 1.x).

- A generator with a numeric `p_set` is *fixed* in `n.optimize()`: clear it (NaN) before an OPF.
- The slack generator needs a numeric `p_set` (e.g. 0) for `n.pf()` to report its output.
- `n.pf()` does not enforce reactive limits: apply PV-PQ switching for planning studies.
- A solved network cannot be copied while its solver model is attached (`n.model.solver_model = None`).
- Lines are in ohm/siemens, transformers in p.u. on their own rating.

13 Limitations and next steps

Teaching networks and the demand model are fictitious; one weather year; perfect foresight; no unit commitment, reserves or forecast errors; static AC checks at selected hours only. Next steps: Renewables.ninja profiles and multi-year weather (2015-2023); N-1 checks on the hourly dispatch; dynamic line ratings; battery degradation; real NIGELEC data where available.

A Cost assumptions

Item	Value	Basis
WACC	8 % real	assumption
Utility PV (Niamey)	800 $/kW, 1.5 % FOM, 25 years	IRENA (2024): global average 758 $/kW in 2023
Battery (Niamey)	250 $/kWh + 150 $/kW, 2 % FOM, 15 years	NREL ATB / BNEF ranges
Thermal fuel	180 $/MWh, 0.70 tCO_2/MWh	HFO/diesel engines
Imports	70 $/MWh	illustrative
Remote PV (Agadez)	1 000 $/kW	small remote site
Remote battery	350 $/kWh + 200 $/kW	small remote site
Diesel	1.00 $/l, 0.28 l/kWh, 20 $/MWh O&M, 0.74 tCO_2/MWh	IPCC 2006 emission factor

B Reproducing a case study

```
python scripts/run_all.py cs07          # re-run one case study (notebook + figures + tables)
python scripts/fetch_renewables_ninja.py --token YOUR_TOKEN   # optional: real ninja PV data
python scripts/run_all.py               # rebuild datasets and re-run everything
```